Parity, polynomial degree, and contact-weighted error costs

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Abstract

We prove a degree lower bound for a polynomial whose sign predicts parity with fixed advantage and whose errors have bounded amplitude cost after weighting by the distance to a coupled contact point. For the mixed quadratic and power cost $\min\{|P(x)|^2/(1+\mathrm{dist}(x,y))^\beta, |P(x)|^{2-\beta}\}$, with $0<\beta\le 1/2$, a coupling with Hamming-distance moments of order $\sqrt{m}$ gives a degree lower bound m^{θ_β}. At $\beta=1/2$, the exponent is $\theta_\beta = 0.719367\ldots$, without a logarithmic loss. The conclusion is stable under approximation and under exceptional sets of source–contact pairs whose mass is exponentially small in the degree. We also give a coupling construction for bounded-density subsets of the cube, a logarithmic norm-ratio refinement of the scalar Bernstein inequality, and a relative random-restriction estimate for homogeneous Walsh polynomials. The contact-cost hypothesis is explicit: establishing it for a particular Bayesian posterior or repair problem is a separate geometric and statistical task.

1 The contact-cost theorem

Let μ_m be uniform probability measure on $\{-1,1\}^m$. Write

$$f(x)=\prod_{i=1}^m x_i, \qquad \partial_i P(x)=\frac{P(x)-P(x^{\oplus i})}{2}, \qquad \Delta=\sum_{i=1}^m \partial_i.$$

Thus $\Delta\chi_S=|S|\chi_S$. All unqualified norms and expectations use μ_m. For $1<t<2$, put

$$s_t = 1+\frac{2}{\pi}\arcsin\left(\frac{2}{t}-1\right).$$

We use the dimension-free scalar Bernstein inequality

$$\|\Delta P\|_t \le C_t u^{s_t}\|P\|_t \qquad (\deg P\le u). \tag{1}$$

This is part of the established theory of polynomial inequalities on the cube [3]. Section 3 supplies the needed version from the imaginary-power estimate of Carbonaro and Dragičević [1].

Fix $0<\beta\le 1/2$ and put $p=2-\beta$. A contact law is a joint distribution π of (U,V) in the cube whose first marginal is exactly μ_m. Set $L=\mathrm{dist}(U,V)$. For a margin-separated function F, meaning $|F|\ge 1$, define

$$E_F=\{x : f(x)F(x)<0\}.$$

Allow an arbitrary exceptional set B of *pairs* (U,V), and define

$$\mathcal{C}_{\pi,B}(F)=\mathbb{E}_\pi\left[\mathbf{1}_{\{U\in E_F,\ (U,V)\notin B\}}\min\left\{\frac{|F(U)|^2}{(L+1)^\beta},\ |F(U)|^p\right\}\right]. \tag{2}$$

In particular, an exceptional set of source points is permitted. No independence between amplitudes, labels, distances, or exceptions is assumed.

Theorem 1.1 (Stable contact-cost calibration). *Fix $c_0 > 0$, $K < \infty$, and $C_* < \infty$. Suppose that:*

1. *$|F| \geq 1$ and $\mathbb{E}[f \operatorname{sign} F] \geq c_0$;*
2. *Q has degree at most $u \geq 1$ and $\|F - Q\|_p \leq e \leq c_0/2$;*
3. *$\pi(B) \leq \delta$ and*
$$\delta^{1/p-1/2}(p-1)^{-u/2} \leq \frac{1}{16}; \tag{3}$$
4. *$\mathcal{C}_{\pi,B}(F) \leq K$ and*
$$\|L+1\|_{L^p(\pi)} \leq C_*\sqrt{m}. \tag{4}$$

Then
$$u \geq c\, m^{\theta_\beta}, \qquad \theta_\beta = \frac{1-\beta/4}{s_p}, \tag{5}$$
where $c > 0$ depends only on the displayed fixed parameters.

The exceptional set may depend arbitrarily on F, Q and the contact law. Its small probability is not sufficient on its own; condition (3) controls its amplitude contribution. For $\delta = \rho^b$, it suffices to have $u \leq a \log(1/\rho)$ with a sufficiently small.

Corollary 1.2 (Explicit contact exponent). *The exponent in Theorem 1.1 is*
$$\theta_\beta = \frac{1-\beta/4}{1+(2/\pi)\arcsin(\beta/(2-\beta))}. \tag{6}$$
For $\beta = 1/2$, it equals $0.7193671500\ldots$. There is no endpoint logarithmic loss: the single exponent $p = 2-\beta$ is used throughout.

1.1 Proof of the calibration theorem

We first record two simple consequences of parity.

Lemma 1.3 (Parity and positivity). *If $|F| \geq 1$, $g = \operatorname{sign} F$, and $\mathbb{E}[fg] \geq c_0$, then*
$$\mathbb{E}[g\Delta F] \geq I(g) \geq mc_0, \qquad I(g) = \sum_i \mathbb{P}\{g(X) \neq g(X^{\oplus i})\}. \tag{7}$$
For every real Q and $1 \leq t < \infty$,
$$\|Q\|_t \leq 2\|(fQ)_-\|_t + \frac{2}{m}\|\Delta Q\|_t. \tag{8}$$

Proof. On an edge where g changes sign, margin separation gives $(g(x)-g(x^{\oplus i}))(F(x)-F(x^{\oplus i})) \geq 4$. Edge symmetrization proves the first inequality in (7). Also $\mathbb{E}[fg] = \mathbb{E}[f\partial_i g]$, so $|\mathbb{E}[fg]| \leq \mathbb{E}|\partial_i g| = \mathrm{Inf}_i(g)$ for every i. Summing proves its second inequality.

Let A be normalized cube adjacency and $q = fQ$. Since parity reverses on every edge, $(I+A)q = 2f\Delta Q/m$. Write its right side as h. Positivity of A gives $q_+ \leq Aq_- + |h|$. Contractivity and the triangle inequality give $\|q\|_t \leq 2\|q_-\|_t + \|h\|_t$. □

Proof of Theorem 1.1. Set $X = \|Q\|_p$, and let

$$b = |F(U)|\mathbf{1}_{\{U\in E_F,\ (U,V)\notin B\}}, \qquad z = \min\{b^2/(L+1)^\beta, b^p\}, \qquad W = \|b\|_{L^p(\pi)}.$$

Pointwise, $b \le z^{1/2}(L+1)^{\beta/2} + z^{1/p}$. Since $2 - p = \beta$, Hölder's inequality with exponents $2/p$ and $2/\beta$ yields

$$\|z^{1/2}(L+1)^{\beta/2}\|_{L^p(\pi)} \le (\mathbb{E}_\pi z)^{1/2}\|L+1\|^{\beta/2}_{L^p(\pi)}, \tag{9}$$

$$W \le K^{1/2}C_*^{\beta/2}m^{\beta/4} + K^{1/p} \le Cm^{\beta/4}. \tag{10}$$

Keeping the quadratic part of the contact cost is essential for this bound.

The exact uniform first marginal and Hölder's inequality give

$$\|Q(U)\mathbf{1}_B\|_{L^p(\pi)} \le \delta^{1/p-1/2}\|Q\|_2 \le \delta^{1/p-1/2}(p-1)^{-u/2}X \le X/16,$$

where the second inequality is the degree-u hypercontractive comparison. It follows that $\|F\mathbf{1}_{E_F}\|_p \le W + X/16 + e$. The negative-part map is 1-Lipschitz, so Lemma 1.3 implies

$$X \le 2W + X/8 + 4e + \frac{2}{m}\|\Delta Q\|_p. \tag{11}$$

If $2C_pu^{s_p}/m \ge 1/4$, then $u \ge cm^{1/s_p}$, which is stronger than the conclusion. Otherwise (1) absorbs the last term in (11), giving $X \le C(W+e)$.

Each ∂_i is an L^1 contraction, so $\|\Delta(F-Q)\|_1 \le me$. By (7),

$$\|\Delta Q\|_p \ge \mathbb{E}[g\Delta Q] \ge m(c_0 - e) \ge mc_0/2.$$

Combining this with (1), (10), and the bound on X proves (5). Formula (6) is the substitution $p = 2 - \beta$. □

2 Balanced contacts with short Hamming distance

The distance assumption has a direct construction when contacts are drawn from a set of fixed positive measure.

Theorem 2.1 (A bounded-density coupling)**.** *If ν is a probability measure on the m-cube with $d\nu/d\mu_m \le C$, there is one coupling $X \sim \mu_m$, $Y \sim \nu$ such that, for every $t \ge 0$,*

$$\mathbb{P}\{\mathrm{dist}(X,Y) > 2\sqrt{m(\log C + 2t)} + t\} \le 2e^{-t}. \tag{12}$$

In particular, for every $q \ge 1$,

$$\|\mathrm{dist}(X,Y)\|_q \le C_0\sqrt{m(\log C + q)} + C_0 q.$$

Proof. Expose coordinates in a fixed order. For the target measure let $p_i(y_{<i}) = \nu(Y_i = 1 \mid Y_{<i} = y_{<i})$, $\delta_i = |p_i - 1/2|$, and $S_i = \sum_{j\le i}\delta_j^2$. Let L_i denote the likelihood ratio on the first i coordinates. On a positive-likelihood prefix,

$$\mathbb{E}_{\mu_m}[\sqrt{L_i} \mid Y_{<i}] = \sqrt{L_{i-1}}\,\frac{\sqrt{p_i} + \sqrt{1-p_i}}{\sqrt{2}}.$$

The square of the last factor is at most $1-\delta_i^2 \le e^{-\delta_i^2}$. Thus $\sqrt{L_i}e^{S_i/2}$ is a nonnegative supermartingale under μ_m. Zero-likelihood prefixes may be assigned arbitrary conditional probabilities. Since $L_m \le C$,

$$\mathbb{E}_\nu e^{S_m/2} = \mathbb{E}_{\mu_m} L_m e^{S_m/2} \le \sqrt{C}\,\mathbb{E}_{\mu_m}\sqrt{L_m}e^{S_m/2} \le \sqrt{C}. \tag{13}$$

Given the full joint past, maximally couple a fresh uniform bit X_i with a bit having probability p_i of being $+1$. The conditional probability of disagreement is δ_i. The two resulting marginals are exactly μ_m and ν. If D is the number of disagreements and $A = \sum_i \delta_i$, the Bernoulli exponential supermartingale gives $\mathbb{E}e^{D-(e-1)A} \le 1$. Since $A \le \sqrt{mS_m}$ and $e-1<2$,

$$\mathbb{P}\{D > 2\sqrt{mS_m} + t\} \le e^{-t}.$$

Equation (13) gives $\mathbb{P}\{S_m > \log C + 2t\} \le e^{-t}$. Their union proves (12); integration proves the moment estimate. □

Two target laws of bounded density can be coupled by gluing their couplings to a common uniform intermediate point. The triangle inequality then gives the same moment scale. In particular, suppose the two sign classes S_+, S_- of a function have measure at least 2ζ, and each contains a contact set A_s of measure at least $\zeta/2$. Couple uniform measure on S_s to uniform measure on A_s, and mix with weights $\mu_m(S_s)$. The source marginal is exactly uniform, the target has the same original sign, and its density is at most $2/\zeta$. For fixed $\zeta > 0$, the distance condition (4) follows.

Remark 2.2 (What the coupling does not prove). A statement about the two marginals does not control a bad relation depending jointly on source and contact. If a geometric repair theorem was proved under a different contact law, it must be proved again for this coupling, or transferred by a valid uniform statistical argument. Likewise, if ζ tends to zero, the factor $\sqrt{\log(1/\zeta)}$ in the distance bound must be retained. The main theorem uses fixed contact mass.

3 A logarithmic Bernstein refinement

The next estimate makes the norm dependence in scalar Bernstein explicit. Its proof follows the fixed-vector interpolation method of Domelevo et al. [2], with strip harmonic measure in place of Gaussian damping. We make no priority claim for the interpolation refinement.

Theorem 3.1. *For $1 < t < 2$, set $\sigma_t = (2/\pi)\arcsin(2/t-1)$. Every nonzero real cube polynomial of degree at most $u \ge 1$ satisfies*

$$\|\Delta P\|_t \le C_t u \left[1 + \log\frac{\|P\|_2}{\|P\|_t}\right]^{\sigma_t} \|P\|_t. \tag{14}$$

The constants are bounded when t ranges over a compact subinterval of $(1,2)$.

Proof. Let $0 < \varepsilon \le (t-1)/t$ and define

$$q = \frac{t(1-\varepsilon)}{1-t\varepsilon/2}, \qquad \frac{1}{t} = \frac{\varepsilon}{2} + \frac{1-\varepsilon}{q}.$$

We first prove

$$\|\Delta P\|_t \le C_t u \varepsilon^{-\sigma_t} \|P\|_2^{\varepsilon}\|P\|_q^{1-\varepsilon}. \tag{15}$$

For $L = \Delta + \gamma I$, $\gamma > 0$, the imaginary-power estimate of [1, Proposition 11] is

$$\|L^{iv}\|_{q\to q} \le C(q^*-1)(1+|v|)^{1/2}e^{\phi_q|v|}, \qquad \phi_q = \arcsin(2/q-1).$$

Its constants are independent of the dimension and of γ. Fix a real g with $\|g\|_{t'} = 1$, put $N = 1/\varepsilon$, and form the holomorphic bilinear pairing

$$G(z) = \mathbb{E}\left[(L^{N(1-z)}P)\, g_z\right], \qquad g_z = \operatorname{sign}(g)|g|^{t'((1-z)/2+z/q')}.$$

The value is set to zero where $g = 0$. At $\theta = 1 - \varepsilon$, it equals $\mathbb{E}[(LP)g]$. On the left boundary, $|G(iv)| \le (u+\gamma)^N \|P\|_2$; on the right,

$$|G(1+iv)| \le C(q^* - 1)(1 + N|v|)^{1/2} e^{\phi_q N|v|} \|P\|_q.$$

On the finite cube G is bounded on the closed strip, so the Poisson inequality applies to $\log|G|$.

The right-boundary harmonic-measure density at θ is

$$w_\varepsilon(v) = \frac{\sin(\pi\varepsilon)}{2(\cosh(\pi v) - \cos(\pi\varepsilon))}, \qquad \int w_\varepsilon = 1 - \varepsilon.$$

The elementary bounds needed here are

$$\int |v| w_\varepsilon(v)\, dv \le \frac{2\varepsilon}{\pi}\big(\log(1/\varepsilon) + 1\big),$$
$$\int \log(1 + |v|/\varepsilon) w_\varepsilon(v)\, dv \le C.$$

For the first, expand

$$\frac{\sin a}{\cosh x - \cos a} = 2\sum_{n\ge 1} \sin(na) e^{-nx}$$

and integrate, obtaining

$$(2/\pi^2)\int_0^{\pi\varepsilon} -\log(2\sin(s/2))\, ds.$$

For the second, use $w_\varepsilon(v) \le \pi\varepsilon/[4(v^2+\varepsilon^2)]$ and rescale.

The left boundary has mass ε. Integrating the boundary logarithms consequently gives

$$|G(\theta)| \le C_t(u+\gamma)\|P\|_2^{\varepsilon}\|P\|_q^{1-\varepsilon}\varepsilon^{-2\phi_q/\pi}.$$

Here $q \ge 1 + (t-1)/2$, and $|\phi_q - \phi_t| \le C_t\varepsilon$, so the last power is at most $C_t\varepsilon^{-\sigma_t}$. Taking the dual supremum and then $\gamma \downarrow 0$ proves (15).

Since $q < t$, $\|P\|_q \le \|P\|_t$. Take $\varepsilon = \min\{(t-1)/t, [1 + \log(\|P\|_2/\|P\|_t)]^{-1}\}$ to obtain (14). All bounds used above are uniform on compact exponent intervals. □

Hypercontractivity gives $\log(\|P\|_2/\|P\|_t) \le (u/2)\log(1/(t-1))$. Substitution recovers (1), since $s_t = 1 + \sigma_t$. Theorem 3.1 therefore does not prove the general linear-degree Bernstein conjecture; it gives linear degree when the norm ratio is bounded.

4 Homogeneous polynomials under random restrictions

For completeness we record a relative restriction estimate that avoids any ambient-dimensional norm loss in a structured case. A Walsh polynomial is homogeneous of degree r if all its nonzero Fourier coefficients have cardinality r.

Theorem 4.1. *Let $1 < t \le 2$, let P be homogeneous of degree r on the k-cube, and choose a uniformly random m-coordinate face, with its outside coordinates uniform. Write P_R for the restriction and $(P_R)_{>u}$ for its Fourier part above degree u. For $j = u + 1$,*

$$\left(\mathbb{E}_R \|(P_R)_{>u}\|_t^t\right)^{1/t} \le \binom{m}{j} B_t^j \left(\frac{r}{k}\right)^{j/2} \|P\|_t, \qquad B_t = (t-1)^{-1/2} 2^{1-t/2}. \tag{16}$$

The left side is zero if $j > r$ or $j > m$. The same result holds with arbitrary additional nuisance variables and homogeneity in the active coordinates. A constant plus one homogeneous level obeys the bound with an additional factor 2.

Proof. Let $\varphi(a) = \operatorname{sign}(a)|a|^{t-1}$. For real a, b,

$$(\varphi(a) - \varphi(b))(a - b) \ge (t-1) \frac{(a-b)^2}{(|a| + |b|)^{2-t}}.$$

Edge symmetrization, Hölder's inequality over the input and a uniform coordinate, and $\Delta P = rP$ give

$$\frac{1}{k} \sum_i \|\partial_i P\|_t^t \le B_t^t \left(\frac{r}{k}\right)^{t/2} \|P\|_t^t. \tag{17}$$

For clarity, the symmetrized energy is $\mathbb{E}[\varphi(P)\Delta P] = (1/4)\sum_i \mathbb{E}[(\varphi(P) - \varphi(P^{\oplus i}))(P - P^{\oplus i})]$. The other Hölder factor is bounded by $\mathbb{E}_i \mathbb{E}(|P| + |P^{\oplus i}|)^t \le 2^t \|P\|_t^t$, which gives exactly the displayed B_t.

For distinct derivatives indexed by S, the polynomial $\chi_S \partial_S P$ is independent of the coordinates in S and homogeneous of degree $r - |S|$ in the remainder. Iteration of (17) therefore yields

$$\mathbb{E}_{|S|=j} \|\partial_S P\|_t^t \le B_t^{jt} \left[\frac{(r)_j}{(k)_j}\right]^{t/2} \|P\|_t^t.$$

Here $(a)_j = a(a-1)\cdots(a-j+1)$.

For any function F on an m-cube there is the exact identity

$$F_{\ge j} = \sum_{|S|=j} \chi_S \, j \int_0^1 (1-z)^{j-1} T_z^{[m]\setminus S}(\chi_S \partial_S F)\, dz.$$

Indeed a Fourier monomial of degree $\ell \ge j$ receives coefficient $\binom{\ell}{j} j \int_0^1 z^{\ell-j}(1-z)^{j-1}\, dz = 1$. Each integral operator is an L^t contraction. Minkowski's inequality on the joint space of the random restriction and its input, followed by the preceding derivative bound and $(r)_j/(k)_j \le (r/k)^j$, proves (16). For a constant plus one homogeneous level, remove the active conditional mean; the remaining norm is at most twice the original norm. □

5 Scope and remaining questions

Theorems 1.1 and 2.1 are independent statements about functions on a finite cube. In applications to kernel or neural-network interpolation, a geometric argument must show that a bounded repair budget controls the cost (2) under the same contact law. If the head or coupling is selected from training data, a uniform statistical argument is also needed. Neither follows from the marginal-density statement alone.

The homogeneous restriction estimate does not extend here to arbitrary sums of Fourier levels. A summable relative restriction bound at effective degree rm/k would imply the general linear

Bernstein inequality: average $\Delta_J P = \sum_{u \geq 0} (P|_J)_{>u}$ over random sets J of size approximately k/r. Thus such an extension is a substantive additional problem. Adaptive normalization on each restricted face can also destroy a direct comparison to the global norm.

The main calibration exponent remains below one. Improving it to a dimension-free obstruction for all degrees $u \leq \alpha m$, with a fixed small $\alpha > 0$, would require an additional relation between low polynomial degree, amplitude distribution, and contact geometry. This note does not establish a full $\exp(ck \log(ed/k))$ posterior sample lower bound.

References

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